INCH-POUND

MIL-PRF-1/148D
23 January 2007
SUPERSEDING
MIL-E-1/148C
12 December 1974

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE C6A

Inactive for new design
after 7 March 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this specification sheet and MIL-PRF-1.

DESCRIPTION: Triode, xenon gas.

See figure 1.

Mounting position: Base down.

Weight: 6 ounces nominal (170.1 grams).

ABSOLUTE RATINGS:

Parameter:	Ef	Ep	epx	epy	Ib	ib	t	tk	TA	F
Unit:	V ac	V ac	v	v	A dc	a	Minutes	sec	°C	Hz
Maximum:	2.75	---	600	300	6.4	77	---	40	70	380
Minimum:	2.25	---	---	---	---	---	---	---	-55	54
Test conditions:	2.5	110	---	---	6.4	---	3	---	---	60 (Note 1)

GENERAL:

Qualification - Not required.

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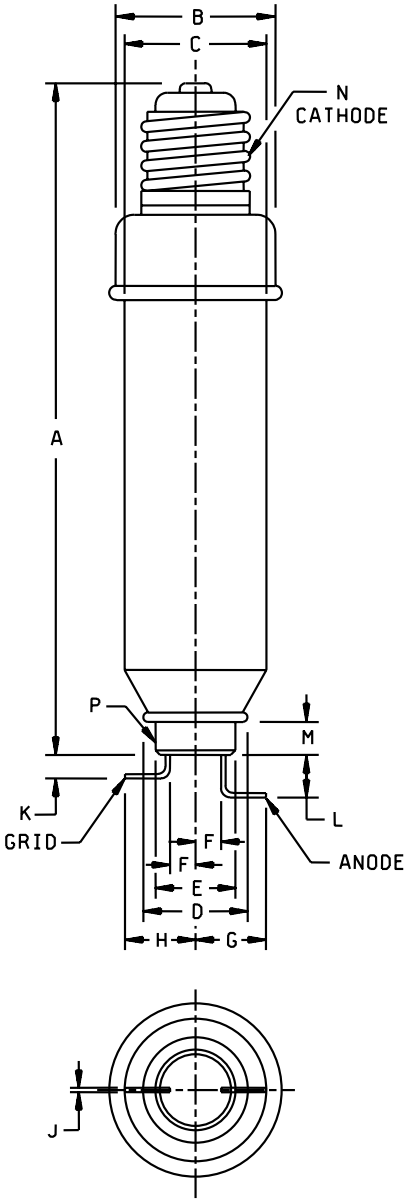

Letter	Dimensions			
	Minimum		Maximum	
	Inches	mm	Inches	mm
Conformance inspection, part 2				
A	10.56	268.22	11.57	293.88
B	2.12	53.85	2.38	60.45
Conformance inspection, part 3 (see note)				
C	1.90	48.26	2.10	53.34
D	---	---	1.63	41.40
E	1.10	27.94	1.14	28.96
F	.26	6.60	.30	7.62
G	1.00	25.40	1.07	27.18
H	.93	23.62	1.00	25.40
J	.09	2.29	.13	3.30
K	.21	5.33	.29	7.37
L	.40	10.16	.47	11.94
M	.37	9.40	.40	10.16
N	Base: Mogul screw, G2-3 (EIA)			
P	Cap: Porcelain with anode prong marked "a" and grid prong "g".			

NOTE: A sample size of three tubes shall be checked annually, with no failures allowed. In case of a sample failure, the test shall become a conformance inspection, part 2 test, for three successful consecutive submissions at which time the test may revert to the conformance inspection, part 3 test.

FIGURE 1. Outline drawing of electron tube type C6A.

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TABLE I. Testing and inspection.

MIL-STD-1311 method	Requirement or test	Notes	Conditions	Acceptance level (see note 10)	Symbol	Limits		Unit
						Min	Max	
	Conformance inspection, part 1							
3206	Operation	---	Rg = 10,000 ohms	0.65	Ecc	1.35	-2.75	V dc
3241	Heater current	---		0.65	If	16	20	A ac
---	AC starting voltage	4	Ecc = 3 V dc; Rg = 1,000 ohms	0.65	Ep	---	43	V ac
---	Average arc drop	5	Ecc = 3 V dc; Rg = 1,000 ohms	0.65	Etd	---	10.5	V dc
---	Peak forward voltage	6	Rg = 1,000 ohms	0.65	Ecc	---	-5.3	V dc
3216	Grid current	---	Rg = 1 MegΩ	0.65	Ic	---	5.0	μA dc
1231	Peak emission by voltage drop	---	Grid connected to anode through 10,000 ohms; ib = 77 a	0.65	etd	---	45	v
	Conformance inspection, part 2							
---	No-load operation	7, 8	Ep = 0; t = 100 hours	---	---	---	---	---
	Conformance inspection, part 3							
---	Life-test	---	Group D; operation; t = 500 hours	---	---	---	---	---
---	Life-test end points:							
1231	Peak emission by voltage drop	---		---	etd	---	67	v
3206	Operation	---		---	Ecc	+1.35	-2.75	V dc
3216	Grid current	---		---	Ic	---	5	μA dc
1031	Low-frequency vibration	2, 9		---	---	---	---	---
1041	Shock	9	Hammer angle = 24°; Ef = 2.5 ± 10%	---	---	---	---	---
---	Cold starting	2, 3, 9		---	---	---	---	---

See notes at top of next page.

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NOTES:

1. The following circuit conditions shall apply for all tests:
   a. Anode and grid circuit returns to center tap of secondary of filament transformer.
   b. Filament transformer terminals phased so that tip of mogul base is positive relative to shell and when anode is positive.
2. Characteristics shall be within specified limits.
3. Apply rated filament and anode voltage simultaneously for a 3-minute period with the load at 1.0 ampere dc and then allow the tube to cool for 15 minutes. This application shall be repeated 500 times.
4. The anode voltage shall be lowered until the tube cuts off. The anode voltage at the instant after the tube cuts off shall be within the limit specified.
5. Measure power loss in the tube with a wattmeter. The power loss divided by the anode current (in amperes, as measured on a dc meter) shall be within the limit specified.
6. The anode voltage shall be increased to 210 V ac, with the anode current held at a value not greater than 6.4 A dc. The grid bias shall be increased in a negative direction until the tube cuts off, and shall be within the limit specified.
7. After no-load operation, tubes shall be within specification limits for operation and grid current. In the operation test, the time of operation shall be a minimum of 30 seconds and a maximum of 120 seconds.
8. The maximum peak forward voltage shall not shift more than 120 volts.
9. This test shall be performed annually. A sample size of three tubes shall be used, with no failures allowed. In case of a sample failure, the test shall become a conformance inspection, part 2 test, for three successful consecutive submissions, at which time the test may revert to the conformance inspection, part 3 test.
10. This specification sheet uses accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

Referenced documents. In addition to MIL-PRF-1, this document references the following: MIL-STD-1311.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2007-007)

Review activities:
Army - AR
Navy - AS, CG, MC, OS, SH
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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